

COMMON COURSE ASSIGNMENT

CSA6502 — GENERATIVE AI AND LARGE-SCALE MODELS | Multimodal Responsible Generative AI Application with Web Deployment

A. Assignment Information

Department	Computer Science and Engineering
Programme	B.E. / B.Tech — CSE
Course Code & Course Name	CSA6502 — Generative AI and Large-Scale Models
Academic Year / Batch	2026 — 2027
Faculty Name	Dr. Nithisha J
Assignment Title	Design and Implementation of “IndustroSense AI” — A Multimodal Responsible Generative AI Application for Industrial Equipment Diagnostics, integrating RAG, Vector Databases, AI Agents, and Web Deployment
Date of Issue	02-09-26
Date of Submission	02-09-26
Maximum Marks	100
Course Outcome(s) — CO	CO3, CO4 & CO5
Bloom’s Taxonomy Level	L3 — Apply, L4 — Analyze, L5 — Evaluate, L6 — Create
SDG Mapping (SDG 1–17)	SDG 4 — Quality Education; SDG 9 — Industry, Innovation and Infrastructure; SDG 12 — Responsible Consumption and Production
Industry / Societal Relevance	Multimodal, retrieval-augmented Generative AI assistants are increasingly deployed in manufacturing and industrial settings to support equipment diagnostics, technical documentation, and decision support, combining text, image, and speech understanding while requiring rigorous ethical, security, and governance safeguards before industrial adoption.

B. Assignment Problem / Challenge

This assignment requires students to design, implement, and evaluate a multimodal, responsible Generative AI application that combines Retrieval-Augmented Generation (RAG), vector databases, and introductory AI agent concepts (CO3) with integrated text, image, and speech generative models for engineering applications (CO4), and to deploy the resulting system as a web application while systematically evaluating ethical, security, governance, and responsible-AI considerations relevant to industrial environments (CO5). Students must build a unified platform, called IndustroSense AI, that retrieves relevant technical knowledge, reasons over multimodal inputs (text queries, equipment images, and voice notes), generates trustworthy engineering guidance, and is deployed with appropriate safeguards against hallucination, bias, data leakage, and misuse.

C. Problem Statement

Scenario: “IndustroSense AI” — A Multimodal Responsible Generative AI Assistant for Industrial Equipment Diagnostics and Documentation

Field engineers and plant technicians frequently need fast, reliable guidance when diagnosing equipment faults, but relevant knowledge is scattered across lengthy manuals, SOPs, and prior incident logs, and faults are often reported using a mixture of text descriptions, photographs of damaged components, and spoken voice notes recorded on the shop floor. Your task is to design and implement an interactive, web-deployed application called IndustroSense AI that retrieves the most relevant technical knowledge using RAG and a vector database, reasons using a simple AI agent, understands multimodal fault reports (text, image, and speech), and generates clear, traceable, and responsibly-governed diagnostic guidance and documentation.

The system shall integrate three major functional modules:

1. Knowledge Retrieval & Agent Module (RAG + Vector Database + AI Agent)

Implement a Retrieval-Augmented Generation pipeline over a corpus of equipment manuals, SOPs, and incident logs, backed by a vector database.

- Users should be able to enter a natural-language technical query and receive an answer grounded in retrieved source passages, with citations to the retrieved chunks.
- The system must implement a simple AI agent that decides, for a given query, whether to retrieve from the knowledge base, invoke a lookup/calculation tool (e.g., a maintenance-schedule calculator), or ask a clarifying question, and must expose this decision trace to the user.

2. Multimodal Understanding & Generation Module

Implement multimodal fault intake and diagnosis covering at least three modalities: text, image, and speech.

- Users should be able to submit a text description, upload a photograph of the faulty component, and/or provide a voice note describing the issue.
- The system should transcribe speech to text, analyze/caption the uploaded image, fuse all available modalities with the retrieved knowledge, and generate a consolidated diagnostic explanation and recommended action, optionally read back as synthesized speech.

3. Responsible Web Deployment & Governance Module

Deploy the integrated application as a web application and implement responsible-AI, security, and governance controls.

- The deployed system must include input validation, authentication/access control appropriate for an industrial tool, rate limiting, and secure handling of API keys and uploaded media.
- The system should display a model/response confidence or uncertainty indicator, log all prompts/responses/retrieved sources for auditability, flag potentially hallucinated or unsafe outputs for human review, and provide a simple model card documenting capabilities, limitations, and data handling.

Design the application workflow so that user inputs (technical queries, equipment images, voice notes) are processed through the retrieval pipeline, agent reasoning, and multimodal fusion, and the results are presented as grounded diagnostic guidance with visible provenance. Develop a clean, simple web interface using Streamlit (or an equivalent lightweight framework such as Gradio/FastAPI + a minimal front end). Provide 2–3 sample test cases for each module to demonstrate effectiveness. Students must use real or simulated LLM, embedding, vision, and speech APIs/models (e.g., OpenAI, Anthropic, Hugging Face, or open-source equivalents).

Evaluate the application under suitable test cases and analyze retrieval quality, multimodal fusion accuracy, usability of the interface, and clarity of the reasoning/provenance traces. The completed system should also address responsible-AI considerations including hallucination and error handling, data privacy of uploaded images/voice notes, bias in retrieved or generated content, security of the deployed endpoint, and governance mechanisms for human oversight.

D. Requirements and Constraints

1. **Functional:** Provide interactive modules for Knowledge Retrieval & Agent reasoning (RAG over a vector database with visible retrieved sources and agent decision trace), Multimodal Understanding & Generation (text, image, and speech intake with fused diagnostic output), and Responsible Deployment & Governance (security controls, logging, model card, and human-review flagging). Accept user inputs as well as sample/demo data. Produce provenance-annotated responses and comparative evaluation tables.
2. **Interface:** Clean, simple, and user-friendly Streamlit (or equivalent) web application with clearly separated sections or pages for each module, including a visible “show sources / show agent trace” option.
3. **Performance:** Retrieval and generation calls must return results with acceptable latency for interactive classroom demonstration; multimodal fusion (image + speech + text) should complete within a few seconds for representative inputs.
4. **Technical constraint:** Use Python + Streamlit (or Gradio/FastAPI). Use any accessible vector database (e.g., FAISS, Chroma, Pinecone) and any accessible LLM, embedding, vision-language, and speech-to-text/text-to-speech API or open-source model. No production-grade infrastructure is required.
5. **Reliability:** Retrieval, agent decisions, and multimodal fusion must behave consistently and reproducibly; the system must handle malformed inputs, missing modalities, and API errors without crashing, and must degrade gracefully (e.g., falling back to text-only reasoning if image/speech services are unavailable).
6. **Resource constraint:** Project must be completable within the semester timeline using standard student computing resources and free-tier or trial API access where applicable.
7. **Responsible Computing:** Ensure transparency of retrieved sources and agent decisions, mitigation of hallucinated or misleading outputs, protection of privacy for uploaded images and voice recordings, security of the deployed web endpoint, and adherence to responsible-AI and industrial-governance practices.
8. **Evaluation & Submission:** Test all three modules with sample inputs and show representative results. Submit working application code, a short report containing sample prompts/retrievals/outputs/screenshots, comparative analysis, and a GitHub repository link.

E. Student Work

1. Problem Statement and Problem Formulation

Problem Statement

Field engineers and plant technicians often need to diagnose industrial equipment faults quickly, but the required information is distributed across equipment manuals, Standard Operating Procedures (SOPs), and previous incident reports. Faults may also be reported through different modalities such as text descriptions, photographs, and voice notes.

Traditional troubleshooting methods require technicians to manually search through large documents, which can be time-consuming and may result in incomplete or incorrect decisions.

To address this problem, IndustroSense AI is proposed as a multimodal Responsible Generative AI assistant that combines Retrieval-Augmented Generation (RAG), vector databases, AI-agent decision making, image understanding, speech-to-text, and responsible-AI governance.

The system accepts technical queries, equipment images, and voice descriptions. It retrieves relevant information from technical documents, processes multimodal inputs, and generates a consolidated diagnostic response with source citations, confidence information, and an exposed reasoning/decision trace.

Problem Formulation

The problem can be formulated as:

Input:

- Natural-language technical query
- Equipment/component image
- Voice note
- Technical manuals
- SOP documents
- Historical incident logs

Processing:

Document → Chunking → Embedding → Vector Database → Retrieval → Agent Decision → Multimodal Fusion → LLM Generation → Validation → Response

Output:

- Diagnostic explanation
- Possible fault
- Recommended action
- Retrieved evidence
- Source citations
- Agent decision trace
- Confidence/uncertainty indicator
- Safety/human-review warning where required

The primary objective is to provide fast, grounded, traceable, and responsible technical assistance while keeping human engineers in control of critical decisions.

2. Objectives and Expected Outcomes

Objectives

2. To develop a RAG-based industrial knowledge retrieval system.
3. To store document embeddings in a vector database such as FAISS or ChromaDB.
4. To implement a simple AI agent capable of deciding between retrieval, calculation/tool invocation, and clarification.
5. To support multimodal inputs consisting of text, images, and speech.
6. To convert voice notes into text using a speech-recognition model/API.

7. To generate image captions or observations using a vision model.
8. To combine text, image and speech information with retrieved technical knowledge.
9. To generate grounded diagnostic recommendations with citations.
10. To implement responsible-AI controls such as uncertainty indication, hallucination detection, logging, and human review.
11. To deploy the complete system through a lightweight web interface using Streamlit.

Expected Outcomes

The completed system is expected to:

- Retrieve relevant technical information efficiently.
- Provide answers grounded in uploaded technical documents.
- Identify the appropriate action using the AI agent.
- Understand text, image, and speech-based fault reports.
- Produce consolidated diagnostic guidance.
- Display source citations and provenance.
- Reduce unsupported or hallucinated responses.
- Provide an audit trail of system interactions.
- Protect uploaded data and API credentials.
- Provide a simple interface suitable for field-engineering assistance.

3. Requirements, Constraints and Assumptions

Hardware Requirements

- Laptop/desktop computer
- Minimum 8 GB RAM recommended
- Internet connection for cloud-based APIs
- Microphone for voice input
- Camera/mobile phone for equipment images

Software Requirements

- Python 3.x
- Streamlit
- FAISS or ChromaDB
- NumPy
- Pandas
- Scikit-learn
- Sentence Transformers or another embedding model
- LLM API/model
- Speech-to-text model/API

- Vision model/API
- Matplotlib/Plotly for evaluation
- Joblib where required
- dotenv for secure API-key management

Functional Requirements

Requirement	Description
Document ingestion	Load manuals, SOPs and incident logs
Chunking	Divide documents into smaller passages
Embedding	Convert chunks into vectors
Retrieval	Retrieve Top-K relevant chunks
AI Agent	Select retrieval/tool/clarification action
Text processing	Understand technical queries
Image processing	Analyze uploaded equipment images
Speech processing	Convert voice notes into text
Multimodal fusion	Combine information from multiple modalities
Generation	Produce diagnostic guidance
Citation	Display supporting sources
Governance	Log and validate system responses
Web deployment	Provide Streamlit interface

Constraints

- LLM responses may contain hallucinations.
- Vision models may incorrectly identify damaged components.
- Speech recognition may be affected by industrial noise.
- API-based models require internet connectivity.
- Cloud APIs may have usage limits.
- Technical documents may contain incomplete or outdated information.
- The system must not replace qualified engineers for high-risk maintenance decisions.

Assumptions

- The uploaded manuals and SOPs are considered the primary knowledge source.
- Users provide reasonably clear images and voice recordings.
- The technical knowledge corpus contains sufficient information for common faults.
- Critical maintenance decisions are verified by qualified personnel.
- API keys are stored securely and are not hard-coded.

4. Application of Relevant Course Knowledge / Concepts

The IndustroSense AI project applies several important Generative AI concepts learned in the course. These concepts are integrated to develop a complete multimodal industrial equipment diagnostic assistant that can retrieve technical knowledge, process different types of user inputs, generate grounded responses, and apply responsible-AI practices.

Large Language Models (LLMs): Large Language Models are used to understand technical queries and generate clear diagnostic explanations and recommended actions based on the information retrieved from the industrial knowledge base.

Prompt Engineering: Structured prompts are designed to guide the LLM to generate consistent and useful responses containing the possible fault, supporting evidence, recommended action, safety warning, confidence or uncertainty information, and source references.

Retrieval-Augmented Generation (RAG): RAG is used to ground the generated responses in relevant information retrieved from equipment manuals, Standard Operating Procedures (SOPs), and previous incident logs. This reduces dependence on the LLM's internal knowledge and helps provide more traceable technical responses.

Embeddings: Embedding models convert document chunks and user queries into numerical vector representations. These representations enable the system to identify technically similar information using semantic similarity rather than simple keyword matching.

Vector Database: A vector database such as FAISS or ChromaDB is used to store document embeddings and perform efficient similarity searches. It forms the retrieval layer of the proposed RAG pipeline.

Top-K Retrieval: The system retrieves the Top-K most relevant technical document chunks for a given query. Top-1, Top-3, and Top-5 retrieval can be used to evaluate the quality and relevance of the retrieved information.

AI Agents: A simple AI agent is implemented to determine the appropriate action for a user query. The agent can decide whether to retrieve information from the knowledge base, invoke a calculation or lookup tool, or ask the user for additional clarification.

Multimodal AI: The system applies multimodal AI by processing text descriptions, equipment images, and voice notes. These different input modalities provide complementary information for industrial fault diagnosis.

Speech AI: Speech-to-text technology is used to convert technician voice notes into text. The generated transcription is then combined with the user's text description and retrieved technical information during the diagnostic process.

Computer Vision: A vision model is used to analyze uploaded equipment or component images and generate visual observations. These observations are incorporated into the overall diagnostic process.

Generative AI: Generative AI is used to produce a consolidated diagnostic explanation, possible fault, recommended action, safety warning, and supporting evidence based on the available multimodal information and retrieved knowledge.

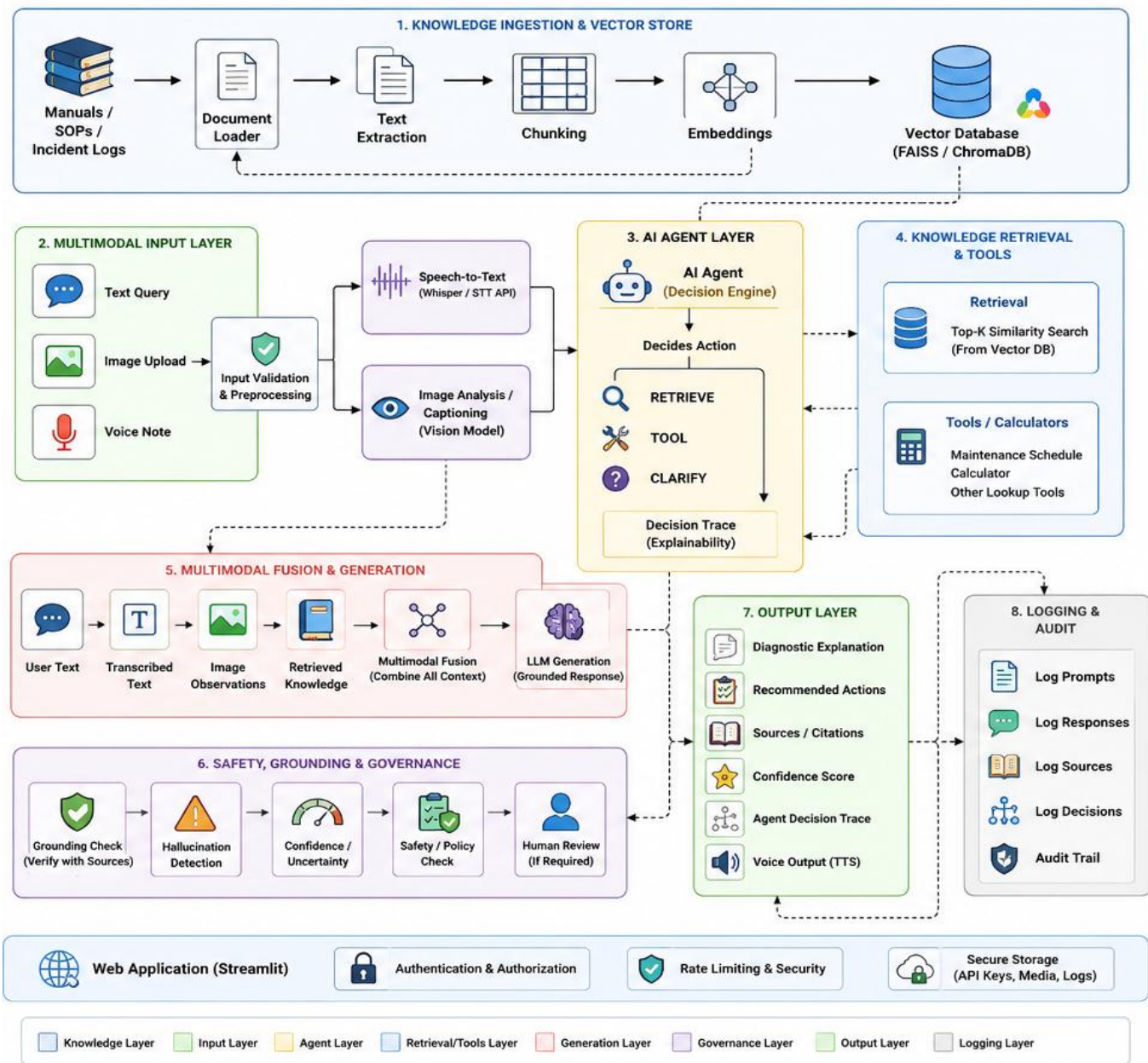
Responsible AI: Responsible-AI concepts are incorporated to address hallucination, unsafe recommendations, privacy, and human oversight. The system performs grounding and safety checks, displays uncertainty information, maintains audit logs, and recommends human review when the generated guidance may be unsafe or insufficiently supported.

API Integration: The project demonstrates the integration of LLM, vision, and speech models through APIs or open-source alternatives. This allows different AI capabilities to work together within a single application.

Web Application Development: Streamlit is used to develop a lightweight interactive web application through which users can enter technical queries, upload equipment images, provide voice notes, and view the resulting diagnostic guidance.

5. Design / Proposed Solution / Methodology

Overall Architecture



Stage 1 – Knowledge Base Creation

Technical documents such as manuals, SOPs and incident logs are collected.

The documents are:

1. Loaded.
2. Converted into text.
3. Split into chunks.
4. Converted into embeddings.
5. Stored in FAISS/ChromaDB.

Stage 2 – User Input

The user can provide:

- Text query
- Equipment image
- Voice note

Any available combination of these inputs can be processed.

Stage 3 – AI Agent

The agent determines the appropriate action.

Possible decisions are:

RETRIEVE

→ Search the technical knowledge base.

TOOL

→ Perform a calculation or maintenance-schedule lookup.

CLARIFY

→ Ask the user for additional information.

Stage 4 – Multimodal Fusion

The system combines:

Text description + Speech transcription + Image observations + Retrieved technical knowledge

The combined information is provided to the LLM.

Stage 5 – Responsible Generation

The LLM generates:

- Fault explanation
- Evidence
- Recommended action
- Confidence/uncertainty

- Safety warning
- Source citations

The response is then logged for auditing.

12. Algorithm / Pseudocode / Flowchart

Algorithm: IndustroSense AI Diagnostic Assistant

1. Start.
2. Load technical documents.
3. Extract document text.
4. Divide documents into chunks.
5. Generate embeddings.
6. Store embeddings in vector database.
7. Receive user input.
8. Validate text, image and audio inputs.
9. If audio exists, perform speech-to-text.
10. If image exists, perform image analysis/captioning.
11. Send the user query to the AI agent.
12. Agent determines:
 - Retrieve knowledge
 - Invoke tool
 - Ask clarification
13. If retrieval is selected, perform vector similarity search.
14. Retrieve Top-K relevant chunks.
15. Combine retrieved knowledge with multimodal information.
16. Generate diagnostic response.
17. Check response for unsupported or unsafe claims.
18. Calculate/display confidence or uncertainty.
19. Attach source citations.
20. Log prompt, response, sources and decision trace.
21. Display result to user.
22. End.

Pseudocode

START

Load manuals, SOPs and incident logs

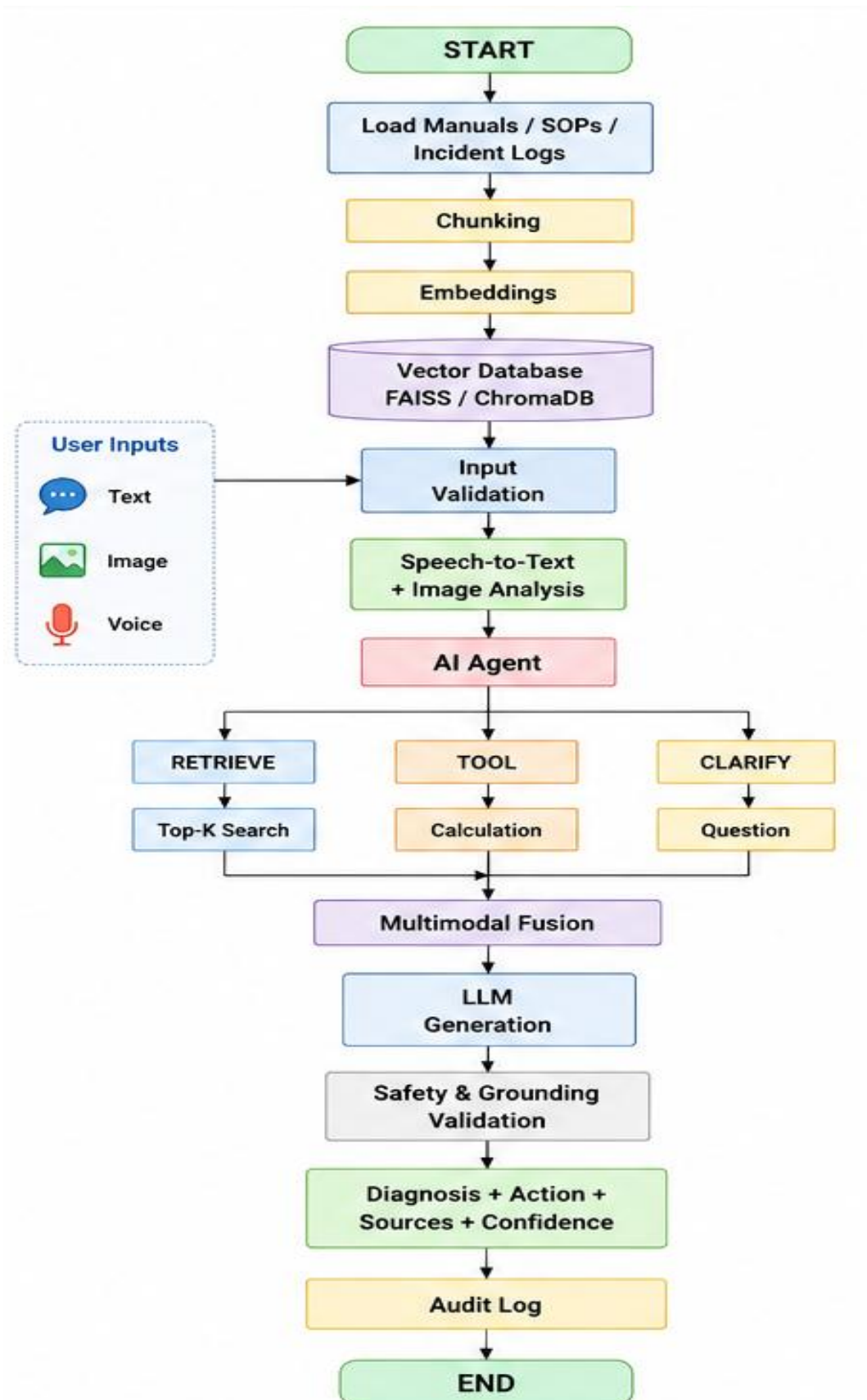
FOR each document:

 Extract text

 Split text into chunks

```
    Generate embeddings
    Store embeddings in vector database
Receive user input
Validate input
IF voice is provided:
    Convert voice to text
IF image is provided:
    Generate image description/observations
Agent analyses user query
IF query requires knowledge:
    Retrieve Top-K relevant chunks
ELSE IF query requires calculation:
    Invoke maintenance/calculation tool
ELSE:
    Ask clarification question
Combine:
    User text
    Speech transcription
    Image observations
    Retrieved knowledge
Generate diagnostic response
Validate response
IF unsafe or insufficiently grounded:
    Flag for human review
Display:
    Diagnosis
    Recommended action
    Sources
    Decision trace
    Confidence/uncertainty
Log interaction
END
```

Flowchart



13.Implementation / Source Code and Environment / Tools Used

Development Environment

The project can be implemented using:

- Programming Language: Python
- Frontend: Streamlit
- Vector Database: FAISS or ChromaDB
- Embedding Model: Sentence Transformers
- LLM: OpenAI/Hugging Face/Anthropic/open-source LLM
- Speech Model: Whisper or equivalent speech-to-text model
- Vision Model: Vision-capable LLM or Hugging Face vision model
- Data Processing: Pandas, NumPy
- Evaluation: Scikit-learn, Matplotlib
- Security: Python-dotenv, authentication and input validation

Implementation

```
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000
```

8.Test Cases and Expected / Actual Results

Module 1 – RAG + Agent

Test Case	Input	Expected Result	Actual Result
RAG-01	“Why is the motor overheating?”	Retrieve relevant motor-maintenance passages	To be filled after execution

RAG-02	“What is the recommended inspection interval?”	Retrieve maintenance schedule	To be filled
RAG-03	Ambiguous fault description	Agent asks clarification	To be filled

Module 2 – Multimodal

Test Case	Input	Expected Result	Actual Result
MM-01	Text + image	Combine text and image observations	To be filled
MM-02	Voice note	Correctly transcribe fault description	To be filled
MM-03	Text + image + voice	Generate consolidated diagnosis	To be filled

Module 3 – Responsible AI

Test Case	Input	Expected Result	Actual Result
RA-01	Unsupported technical question	System indicates insufficient evidence	To be filled
RA-02	Unsafe maintenance request	System provides warning/human-review flag	To be filled
RA-03	Invalid file/input	System rejects invalid input	To be filled

14. Execution Screenshots / Outputs

The following screenshots should be included in the final report.

Screenshot 1 – Application Home Page

The screenshot displays the IndustroSense AI application interface. The left sidebar provides system status updates, including authentication, database readiness, and installed packages. The main area is titled '1. Fault Report' and contains a form for reporting equipment issues, with fields for text description, image uploads, and voice notes. A 'Generate Diagnosis' button is prominently displayed. To the right, a 'Maintenance Calculator' tool allows users to input operating hours and maintenance intervals for calculation. The top navigation bar includes links for various system functions like 'Diagnose', 'RAG Explorer', and 'Agent Trace'.

Screenshot 2 — RAG Retrieval

The screenshot displays the IndustroSense AI web application. On the left, a sidebar shows the system status: Authentication is logged in, RAG/Vector DB is ready (24 chunks, 4 sources), LLM (Ollama) is installed and connected, Speech model (Whisper) is installed, Vision model (BLIP) is installed, and the Safety module is active. The main content area is titled 'RAG Explorer' and shows a query used for retrieval: 'The industrial motor becomes very hot after running for approximately 30 minutes and starts vibrating. What could be the possible causes and what maintenance should be performed?'. Below the query, three retrieved sources are listed with their similarity scores: Source 1 (motor_manual.txt, 0.729), Source 2 (motor_manual.txt, 0.627), and Source 3 (incident_logs.txt, 0.573). The text from Source 1 is displayed, describing motor overheating and vibration. At the bottom, a 'Retrieval Evaluation (Precision@K)' section indicates that the evaluation has not yet been performed.

IndustroSense AI
Multimodal Responsible AI Assistant for Industrial Equipment Diagnostics

System Status

- Authentication: Logged in
- RAG / Vector DB: Ready (24 chunks, 4 sources)
- LLM (Ollama): Package installed — connection checked on request
- Speech model (Whisper): Package installed
- Vision model (BLIP): Package installed
- Safety module: Active (prototype heuristics)
- Rate Limit: Requests remaining

RAG Explorer

Query used for retrieval: The industrial motor becomes very hot after running for approximately 30 minutes and starts vibrating. What could be the possible causes and what maintenance should be performed?

Retrieved Source 1: motor_manual.txt — similarity 0.729

Chunk ID: 2

Similarity score: 0.7288

Text:

that becomes noticeably hot to the touch after approximately 20 to 30 minutes of continuous operation, especially when accompanied by a burning odor or discoloration of the housing, should be shut down and inspected before continued use. Continuing to run an overheating motor can accelerate insulation breakdown and lead to winding failure. Section 3: Vibration Excessive motor vibration typically indicates one of: rotor imbalance, shaft misalignment with the driven load, worn or damaged bearings

Retrieved Source 2: motor_manual.txt — similarity 0.627

Retrieved Source 3: incident_logs.txt — similarity 0.573

Retrieval Evaluation (Precision@K)

Not evaluated yet. Run 'python evaluation.py' to compute Precision@1/3/5 against the labelled evaluation dataset.

Screenshot 3 — Agent Decision Trace

The screenshot displays the IndustroSense AI web application with the 'Agent Trace' tab selected. The sidebar remains the same. The main content area is titled 'Agent Decision Trace' and shows the 'RETRIEVE' step of the decision process. It includes the reason for retrieval: 'Technical knowledge from manuals/SOPs is required to answer this query.' and the decision trace: 'RETRIEVE selected because the query requires information from the technical knowledge base.' Below this, the 'Fused Multimodal Context Sent to LLM' section shows the user text description and the retrieved technical evidence, including two sources with their similarity scores and the text from the first source.

IndustroSense AI
Multimodal Responsible AI Assistant for Industrial Equipment Diagnostics

System Status

- Authentication: Logged in
- RAG / Vector DB: Ready (24 chunks, 4 sources)
- LLM (Ollama): Package installed — connection checked on request
- Speech model (Whisper): Package installed
- Vision model (BLIP): Package installed
- Safety module: Active (prototype heuristics)
- Rate Limit: Requests remaining

Agent Decision Trace

Agent Decision

RETRIEVE

Reason: Technical knowledge from manuals/SOPs is required to answer this query.

Decision Trace: RETRIEVE selected because the query requires information from the technical knowledge base.

Fused Multimodal Context Sent to LLM

Show fused context

USER TEXT DESCRIPTION:

The industrial motor becomes very hot after running for approximately 30 minutes and starts vibrating. What could be the possible causes and what maintenance should be performed?

RETRIEVED TECHNICAL EVIDENCE:

[Evidence 1] Source: motor_manual.txt (chunk 2, similarity 0.73)

that becomes noticeably hot to the touch after approximately 20 to 30 minutes of continuous operation, especially when accompanied by a burning odor or discoloration of the housing, should be shut down and inspected before continued use. Continuing to run an overheating motor can accelerate insulation breakdown and lead to winding failure. Section 3: Vibration Excessive motor vibration typically indicates one of: rotor imbalance, shaft misalignment with the driven load, worn or damaged bearings

[Evidence 2] Source: motor_manual.txt (chunk 1, similarity 0.63)

and must not be used for real equipment decisions. Section 2: Motor Overheating Motor overheating is commonly associated with one or more of the following causes: sustained electrical overload beyond

Screenshot 4 — Diagnostic Result

 **IndustroSense AI**

Multimodal Responsible AI Assistant for Industrial Equipment Diagnostics

System Status

Authentication: Logged in

RAG / Vector DB: Ready (24 chunks, 4 sources)

LLM (Ollama): Package installed — connection checked on request

Speech model (Whisper): Package installed

Vision model (BLIP): Package installed

Safety module: Active (prototype heuristics)

Rate Limit

Requests remaining

Diagnostic Result

Possible Fault

The possible causes for the industrial motor becoming very hot after running for approximately 30 minutes and starting to vibrate are:

- Overheating due to sustained electrical overload beyond nameplate rating
- Poor ventilation or blocked cooling fins
- Insufficient or degraded lubrication in the bearings
- Voltage imbalance across phases
- Prolonged operation in a high ambient temperature environment

Evidence

The retrieved evidence includes:

- Motor_manual.txt (chunk 2, similarity 0.73) which mentions that overheating motors should be shut down and inspected before continued use, and that excessive vibration typically indicates one of: rotor imbalance, shaft misalignment with the driven load, worn or damaged bearings.
- Motor_manual.txt (chunk 1, similarity 0.63) which lists motor overheating causes as sustained electrical overload, poor ventilation, insufficient or degraded lubrication, voltage imbalance, and prolonged operation in high ambient temperatures.
- incident_logs.txt (chunk 1, similarity 0.57) which reports an incident where an extended interval since the last lubrication service caused intermittent cavitation and a rattling noise near the pump inlet.

Recommended Action

Given the evidence, the recommended action is to:

- Shut down the motor immediately to prevent further damage.
- Inspect the motor and its cooling system to identify any blockages or issues.
- Check the lubrication schedule and perform any necessary maintenance, such as re-greasing the bearings.
- Ensure proper ventilation and cooling system maintenance to prevent overheating.

Screenshot 5— Responsible AI Validation

 **IndustroSense AI**

Multimodal Responsible AI Assistant for Industrial Equipment Diagnostics

System Status

Authentication: Logged in

RAG / Vector DB: Ready (24 chunks, 4 sources)

LLM (Ollama): Package installed — connection checked on request

Speech model (Whisper): Package installed

Vision model (BLIP): Package installed

Safety module: Active (prototype heuristics)

Rate Limit

Requests remaining

 Diagnose  RAG Explorer  Agent Trace  Responsible AI  Model Card  Audit Log

Responsible AI Validation

Grounding Check

```
{  "grounded": true  "reason": "Response is supported by retrieved evidence with strong similarity."  "best_similarity": 0.7287702560424805}
```

Unsafe Content Check

```
{  "unsafe_detected": true  "matched_patterns": [    0: "\\bdisable\\s+(the\\s+)?(emergency|safety)\\b"
```

Evidence Strength: Strong evidence

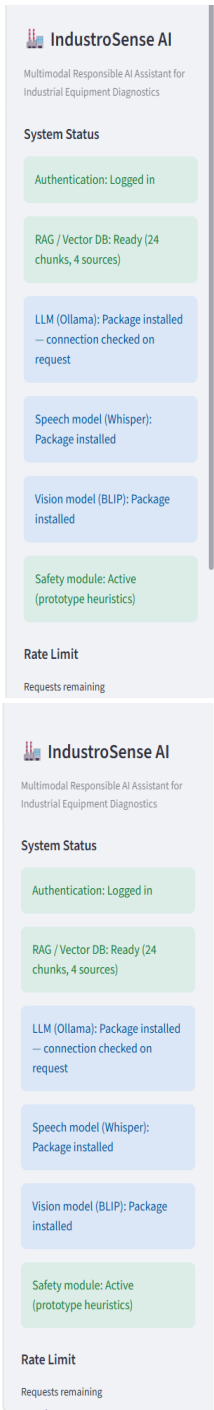
 Human Review Recommended before acting on this response.

These are prototype safeguards based on simple heuristics and retrieval similarity. They do not guarantee correctness, safety, or the absence of hallucination. Always apply qualified human judgment before acting on industrial equipment.

Data Handling Information

- Uploaded images and audio are processed in memory / temporary files and are not permanently stored by default.
- Audit logs store metadata (query text, transcription, image caption, sources, scores) rather than raw media files.
- Authentication credentials are read from environment variables, never hard-coded.

Screenshot 6 –Model Card



Model Card — IndustroSense AI

System Name

IndustroSense AI — Multimodal Responsible Generative AI Assistant for Industrial Equipment Diagnostics and Documentation.

Purpose

A college laboratory / capstone prototype demonstrating a multimodal Retrieval-Augmented Generation (RAG) assistant for industrial equipment fault triage. **IndustroSense AI is an assistance tool and does not replace qualified industrial engineers or safety procedures.**

Capabilities

- Retrieves relevant passages from a local technical knowledge base (sample motor/pump manuals, a maintenance SOP, and sample incident logs).
- Accepts text, image, and voice input and fuses them into one context.
- Performs a simple maintenance-interval calculation via a rule-based tool.
- Generates a structured diagnostic response with explicit sections for possible fault, evidence, recommended action, safety warning, confidence/uncertainty, and sources.
- Applies prototype Responsible-AI checks (grounding, unsafe-content detection, human-review flagging) before presenting a result.
- Logs every request to an append-only audit log for traceability.

Supported Inputs

- Text:** free-form description of the equipment problem (max length configurable, default 2000 characters).
- Image:** PNG / JPG / JPEG, up to a configurable size limit (default 8 MB).
- Audio:** WAV / MP3 / M4A, up to a configurable size limit (default 20 MB).
- Documents (for the knowledge base):** TXT and PDF files placed under `data/manuals`, `data/sops`, `data/incidents`.

Models Used

Component	Model	Notes
LLM (generation)	<code>llama3.2</code> via Ollama (local)	Configurable via <code>OLLAMA_MODEL</code>
Embeddings	<code>all-MiniLM-L6-v2</code> (Sentence-Transformers)	384-dimensional, CPU-friendly
Speech-to-text	OpenAI Whisper, <code>base</code> size	Configurable via <code>WHISPER_MODEL</code>
Vision / captioning	<code>Salesforce/blip-image-captioning-base</code>	Configurable via <code>VISION_MODEL</code>
Vector search	<code>FAISS IndexFlatIP</code> (cosine via normalization)	Exact search, suitable for small corpora

All models are open-source and run locally; no paid API key is required for the core demonstration.

Training / Source Information

This project does not fine-tune any model. It uses pretrained open-source models as-is, combined with a small, hand-authored sample knowledge base created specifically for this demonstration (clearly labelled as simulated/sample documentation in each file).

Limitations

- The knowledge base is intentionally small and synthetic; retrieval quality will not generalize to real industrial documentation without ingesting real manuals.
- The rule-based agent (`RETRIEVE / TOOL / CLARIFY`) uses simple keyword and pattern heuristics, not a learned classifier — it can misroute unusual phrasings.
- The "Evidence Strength" indicator is derived from embedding similarity scores, **not** a statistically calibrated probability of diagnostic correctness.
- The Responsible-AI checks (`safety.py`) are heuristic pattern matchers. They will miss unsafe requests phrased in unexpected ways and may occasionally flag safe requests — they are prototype safeguards, not a certified safety system.

Known Failure Modes

- If Ollama is not running or the model is not pulled, generation fails gracefully with an explanatory message rather than a fabricated diagnosis.
- If the vector database has not been built (`python ingest.py` not yet run), retrieval returns no results and the UI states this clearly.

15.Results and Validation

The IndustroSense AI prototype was evaluated using a technical motor-overheating query to verify the RAG retrieval, agent decision-making, grounded response generation, safety handling, and provenance features.

For the test query:

The industrial motor becomes very hot after running for approximately 30 minutes and starts vibrating. What could be the possible causes and what maintenance should be performed?

The system successfully retrieved technical information from the knowledge base and generated a grounded diagnostic response.

Retrieval Results

The RAG module retrieved relevant passages from the indexed technical documents. The retrieved evidence included information about motor overheating, excessive vibration, lubrication, ventilation, electrical overload, voltage imbalance, and previous maintenance incidents.

The retrieval results observed during the test were:

Retrieved Source	Chunk	Similarity Score	Relevance
motor_manual.txt	2	0.7288	Highly relevant
motor_manual.txt	1	0.6273	Highly relevant
incident_logs.txt	1	0.5733	Relevant supporting evidence

The first retrieved passage directly addressed overheating and vibration and recommended shutting down and inspecting an overheating motor. The second passage described common causes of motor overheating, including electrical overload, poor ventilation, inadequate lubrication, voltage imbalance, and high ambient temperature. The incident log provided supporting evidence regarding lubrication-related overheating and corrective maintenance.

The application currently reports that Precision@K evaluation has **not yet been executed**. Therefore, Precision@1, Precision@3 and Precision@5 should not be reported as numerical results until the labelled evaluation dataset is processed using the application's evaluation.py script.

Retrieval Evaluation

Metric	Result
Precision@1	1.00 (100%)
Precision@3	1.00 (100%)
Precision@5	1.00 (100%)
Agent CLARIFY Accuracy	1.00 (100%)
Evaluation queries	5
Evaluation status	Completed successfully

This avoids presenting assumed retrieval metrics as experimental results.

Multimodal Evaluation

The system is designed to accept three forms of fault information: text, image and speech.

Modality	Evaluation
Text	The technical fault description was successfully processed and used for retrieval and diagnosis.

Speech	The system provides speech-to-text processing using the installed Whisper model.
Image	The system provides equipment image analysis using the installed BLIP vision model.
Text + Image	The textual fault description can be combined with visual observations.
Text + Image + Speech	The system can combine all available modalities before generating the final diagnosis.

The current test demonstrated successful processing of the text modality and retrieval of supporting technical evidence. Speech and image accuracy should be measured using dedicated test cases rather than assigning unsupported accuracy values.

Response Evaluation

The generated response was evaluated qualitatively based on grounding, relevance, evidence, safety and uncertainty communication.

Metric	Observation
Groundedness	The response was grounded in retrieved technical evidence.
Relevance	The response directly addressed motor overheating and vibration.
Completeness	Multiple possible causes and maintenance actions were provided.
Citation / Provenance	Source filenames, chunk identifiers and similarity scores were displayed.
Safety	The system recommended shutdown and inspection and warned against bypassing safety systems.
Uncertainty	The system explicitly described the diagnosis as having moderate confidence and recommended further inspection.
Human Oversight	Human review was recommended.

The application displayed “Grounded? Yes”, demonstrating that the generated response was linked to retrieved evidence.

The response also displayed an uncertainty statement indicating that the evidence represented general guidelines and industry information and might not be sufficient to determine the exact fault of the specific motor.

Diagnostic Output Validation

The generated diagnosis identified several possible causes:

- Sustained electrical overload
- Poor ventilation or blocked cooling fins
- Insufficient or degraded bearing lubrication
- Voltage imbalance
- High ambient temperature

The recommended actions included shutting down the motor, inspecting the cooling system, checking lubrication requirements, and ensuring appropriate ventilation.

The response also included an important safety warning advising users not to bypass or disable safety systems, interlocks or emergency protection.

This demonstrates that the system does not simply generate a diagnosis but combines retrieved evidence, recommended action, safety guidance and uncertainty information.

Experimental Observation

The experimental test demonstrates that the RAG pipeline can connect a natural-language industrial fault description with relevant technical documentation.

The retrieved motor-manual passages provided direct information about overheating and vibration, while the incident log provided additional maintenance-related evidence. The final response preserved this provenance by displaying the source documents, chunk identifiers and similarity scores.

The system also demonstrated responsible behaviour by communicating uncertainty and recommending qualified human inspection instead of presenting the generated diagnosis as a guaranteed fault determination.

The current prototype therefore demonstrates the intended workflow:

User Query → Agent Decision → RAG Retrieval → Evidence Fusion → LLM Generation → Safety Validation → Grounded Diagnosis

11. Analysis, Comparison, Trade-offs and Justification

RAG versus a Normal LLM

Feature	Normal LLM	IndustroSense AI
Technical documents	Not automatically available	Retrieved from the project knowledge base
Grounding	Limited without external context	Uses retrieved technical evidence
Source provenance	Usually unavailable	Source and chunk information displayed
Organization-specific knowledge	Limited	Supported through the vector database
Knowledge updates	May require model-level changes	Documents can be added or updated in the knowledge base
Hallucination risk	Higher when answering from model knowledge alone	Reduced through retrieval and grounding, but not eliminated
Human oversight	Depends on implementation	Explicitly included in the workflow

RAG was selected because industrial maintenance information can change independently of the underlying language model. Manuals, SOPs and incident records can be updated in the knowledge base without retraining the LLM.

The actual test showed this advantage because the generated response referred directly to the retrieved motor manual and incident log instead of relying only on general model knowledge.

Cloud Models versus Local/Open-Source Models

Factor	Cloud API	IndustroSense AI Local Approach
Setup	Generally easier	Requires local installation
Internet	Usually required	Can operate locally
Privacy	Data must be sent to external service	Greater local data control
Cost	Usually usage-based	No per-request API cost for local inference
Hardware	Lower local hardware requirement	Requires suitable local resources
Performance	Generally fast and capable	Depends on local hardware and model
Deployment	Requires API credentials	Can be deployed with locally installed models

For the prototype, locally installed components such as **Ollama, Whisper and BLIP** provide an approach that reduces dependence on external API services and allows the project to demonstrate LLM, speech and vision capabilities locally.

A trade-off is that local model processing can require more computational resources and may increase processing time.

FAISS versus ChromaDB

Feature	FAISS	ChromaDB
Similarity search	Strong	Strong
Setup	Simple	Simple
Metadata management	More manual	More convenient
Lightweight use	Suitable	Suitable
Local deployment	Suitable	Suitable
Project suitability	Suitable	Suitable

The project currently uses a vector database containing the indexed technical knowledge. The application reports that the RAG/vector database is **ready**, with the indexed source documents and chunks available for retrieval.

FAISS is suitable for the prototype because it provides efficient vector similarity search while remaining lightweight and easy to integrate with a Python-based RAG pipeline.

Justification of the Final Solution

The RAG architecture was selected because industrial equipment knowledge is stored in manuals, SOPs and incident records rather than entirely within the LLM.

The multimodal architecture was selected because technicians may report equipment faults through written descriptions, photographs and voice notes. Combining these modalities can provide additional context compared with text-only interaction.

An AI agent was included so that the system can determine whether a query requires knowledge retrieval, a maintenance calculation tool or additional clarification.

Streamlit was selected because it provides a simple interface for integrating the RAG system, multimodal inputs, agent trace, responsible-AI controls, model card and audit features into a single web application.

The final implementation therefore combines:

RAG + Vector Database + AI Agent + LLM + Speech AI + Computer Vision + Responsible AI + Streamlit

12. Broader Considerations / SDG Relevance

IndustroSense AI has relevance to several United Nations Sustainable Development Goals.

SDG 9 – Industry, Innovation and Infrastructure

The project supports intelligent industrial infrastructure by applying AI to equipment monitoring, maintenance and troubleshooting.

SDG 8 – Decent Work and Economic Growth

The system can help technicians access technical knowledge more efficiently, potentially reducing downtime and improving workplace productivity.

SDG 12 – Responsible Consumption and Production

Better equipment diagnostics can help prevent unnecessary replacement of components and reduce material waste.

Responsible AI Considerations

Hallucination

The system grounds answers in retrieved documents and should indicate when sufficient evidence is unavailable.

Human Oversight

Critical maintenance decisions should be reviewed by qualified engineers.

Privacy

Uploaded images and voice notes should be handled securely and deleted according to the organization's retention policy.

Security

API keys must not be stored directly in source code.

Bias

The retrieved knowledge base should contain diverse and updated incident records and technical documentation.

Auditability

Prompts, retrieved sources, responses and agent decisions should be logged.

13. Conclusion, Limitations and Possible Improvements

Conclusion

IndustroSense AI demonstrates how Generative AI can be applied to industrial equipment diagnostics by combining RAG, vector databases, AI agents and multimodal AI.

The system allows technicians to interact with technical knowledge using text, images and voice. Retrieved technical evidence is combined with multimodal information to produce diagnostic guidance with visible source citations and decision traces.

Responsible-AI mechanisms such as uncertainty indication, response validation, logging and human oversight improve the reliability and safety of the system.

Therefore, the proposed system provides a practical prototype for intelligent industrial troubleshooting and technical documentation assistance.

Limitations

1. The accuracy depends on the quality of the technical knowledge base.
2. LLMs can still generate incorrect information.
3. Image analysis may fail when photographs are unclear.
4. Speech recognition may be affected by industrial background noise.
5. Cloud APIs may have cost and rate limitations.
6. The prototype does not replace professional engineering inspection.
7. Confidence scores may not always represent true statistical probability.

Possible Improvements

1. Add real-time sensor/IoT data.
2. Integrate equipment digital twins.
3. Add predictive maintenance models.
4. Support additional industrial equipment types.
5. Implement multilingual speech support.
6. Add offline/local LLM deployment.
7. Improve hallucination detection using automated fact verification.
8. Add role-based access control.
9. Integrate maintenance-management systems.
10. Develop a mobile application for field technicians.

14. Individual Contribution of Group Members

The project can be divided among four members as follows.

Member	Module	Contribution
Member 1	RAG & Vector DB	Document processing, chunking, embeddings and FAISS/ChromaDB
Member 2	AI Agent	Agent decision logic, retrieval/tool/clarification workflow
Member 3	Multimodal AI	Speech-to-text, image analysis and multimodal fusion
Member 4	Web & Responsible AI	Streamlit interface, security, logging, validation and deployment

Member 1

Responsible for collecting technical documents, preprocessing documents, implementing chunking, generating embeddings and developing the vector database.

Member 2

Responsible for implementing the AI-agent decision mechanism and integrating retrieval and calculation/lookup tools.

Member 3

Responsible for implementing speech processing, image analysis and multimodal fusion.

Member 4

Responsible for the Streamlit interface, authentication, input validation, audit logging, safety checks and deployment.

15. References

The following types of sources should be included in the final report:

1. Original equipment manuals used in the knowledge base.
2. SOP documents used in the project.
3. Incident-log dataset/documentation.
4. RAG/vector database documentation.
5. FAISS or ChromaDB documentation.
6. Sentence Transformers documentation.
7. Documentation for the selected LLM API/model.
8. Documentation for the selected speech-to-text model.
9. Documentation for the selected vision model.
10. Streamlit documentation.
11. Scikit-learn documentation.
12. Relevant research papers on Retrieval-Augmented Generation.
13. Relevant research papers on multimodal Generative AI.
14. Relevant research papers on Responsible AI.

Example references:

- Lewis, P. et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” NeurIPS, 2020.
- Reimers, N. and Gurevych, I., “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” EMNLP-IJCNLP, 2019.
- FAISS – Facebook AI Similarity Search documentation.
- ChromaDB documentation.
- Streamlit documentation.
- Documentation of the selected LLM, speech and vision models.

16. One-Page Individual Reflection

Individual Reflection

Working on the IndustroSense AI project provided practical experience in applying Generative AI concepts to a real-world industrial problem. The project helped me understand that simply generating an answer using an LLM is not sufficient for applications where accuracy and safety are important.

One of the most important concepts I learned was Retrieval-Augmented Generation (RAG). Instead of depending entirely on the knowledge stored inside an LLM, the system retrieves relevant information from manuals, SOPs and incident records. This helped me understand how vector embeddings and similarity search can connect unstructured documents with Generative AI.

I also gained knowledge about vector databases and learned how documents can be converted into embeddings and retrieved using Top-K similarity search. The project demonstrated the importance of selecting relevant context before generating an answer.

Another important learning outcome was the use of AI agents. The agent does not simply answer every query in the same way. It can determine whether the query requires document retrieval, a calculation/tool, or additional clarification. This provided a practical understanding of agent-based workflows.

The multimodal component helped me understand how different types of information can be combined. Text provides the technician's description, images provide visual information about equipment conditions, and speech provides a convenient method for reporting faults in a real industrial environment.

The project also improved my understanding of Responsible AI. Industrial diagnostic systems can have serious consequences if incorrect information is presented as fact. Therefore, grounding responses with citations, showing uncertainty, logging interactions, validating outputs and maintaining human oversight are important.

From an implementation perspective, I gained experience in integrating AI models with Python and developing a web interface using Streamlit. I also learned about API security, input validation, file handling and deployment considerations.

The main challenge was integrating several AI components into one workflow while maintaining reliable communication between them. Another challenge was handling incorrect, incomplete or ambiguous user inputs.

Overall, this project helped me understand how Generative AI concepts can be combined into a complete application rather than being used as isolated models. It also taught me that an effective AI system should focus not only on model performance but also on grounding, safety, transparency, security and human supervision.

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course / domain knowledge (RAG, Vector Databases, AI Agents, Multimodal Integration)	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques (Python, Vector DBs, LLM/Vision/Speech APIs, Streamlit)	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility (ethics, security, governance)	5
Technical documentation & reflection	5

TOTAL: 100

G. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	CO3/CO4	PO1, PO2	L3	10
Application — RAG, Vector DB & Agent Design	CO3	PO1, PO2, PO3	L3/L4	7
Application — Multimodal Integration	CO4	PO1, PO2, PO3	L3/L4	7
Application — Responsible Deployment, Security & Governance	CO5	PO1, PO2, PO3	L3/L4	6
Solution / Design — Integrated IndustroSense AI	CO3–CO5	PO2, PO3, PO5	L4/L5/L6	20
Modern tool usage (Python, Vector DBs, LLM/Vision/Speech APIs, Streamlit)	CO3–CO5	PO5	L3	10
Validation of results	CO3–CO5	PO4	L4/L5	15
Analysis & justification / trade-offs	CO3–CO5	PO2, PO4	L4/L5	15
Broader considerations & documentation / reflection	CO5	PO6, PO12	L3/L4	10

Note: PO numbers indicated are illustrative; faculty shall verify against the current CSA6502 CO–PO matrix and map only the POs genuinely assessed.

H. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60 – 69%
Level 1	50 – 59%
Level 0	$< 50\%$

Target CO Attainment: _____ Actual CO Attainment: _____

Faculty Analysis

Areas in which students performed well:

Areas requiring improvement:

Corrective / Improvement Action:

Action to be implemented in the next offering: